Election Process Education Agent — Implementation Prompt Document for Antigravity Agent Platform

Executive Summary

This document defines the complete product, technical, design, safety, and implementation specification for building an **Election Process Education Agent** on **Antigravity's Agent platform**. The product is a multilingual civic-information assistant for India that helps users understand:

- voter registration processes
- eligibility basics
- voter ID and document requirements
- polling day procedures
- accessibility support for voters
- election-related FAQs
- where to find official information
- how to recognize and avoid misinformation

The agent is strictly **educational and informational**. It must **not**:

- persuade users to vote for or against any candidate, party, or ideology
- provide political targeting or campaign guidance
- manipulate turnout or voting behavior
- provide legal advice
- claim certainty when official sources are unclear
- store sensitive voter data unnecessarily

The system should support **22 scheduled Indian languages**, offer **text and voice interaction**, and prioritize official source grounding via the **Election Commission of India (ECI)** and official state Chief Electoral Officer (CEO) websites.

This specification is designed to be implementation-ready for product, engineering, design, QA, and DevOps teams.

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1. Project Requirements and Scope

1.1 Product Vision

Build a trustworthy, multilingual, mobile-first civic-information agent that helps people in India understand the election process in simple language, across text and voice, using official sources and safe educational guidance.

The agent should reduce confusion, improve access to election process knowledge, and direct users to official channels whenever personalized or time-sensitive verification is required.

1.2 Problem Statement

Many voters face challenges such as:

- confusion about voter registration steps
- uncertainty about eligibility
- difficulty finding state-specific election procedures
- limited access to clear multilingual guidance
- misinformation spread through social media and informal networks
- barriers for first-time voters, rural users, and low-digital-literacy users
- lack of accessible information for persons with disabilities

The product solves for **clarity**, **discoverability**, **translation**, **voice access**, and **official-source routing**.

1.3 Goals

Primary Goals

- Explain the Indian election process clearly in plain language.
- Support multilingual educational interactions in 22 scheduled Indian languages.

- Provide step-by-step guidance for:
  - voter registration
  - checking eligibility criteria
  - understanding required documents
  - polling day process
  - finding official sources
- Minimize misinformation by grounding responses in curated official content.
- Offer voice input/output for low-literacy and mobile users.
- Maintain non-partisan, non-political, non-manipulative behavior.

Secondary Goals

- Reduce user drop-off when seeking official election information.
- Improve user confidence in understanding procedural steps.
- Provide state-specific routing where relevant.
- Capture user feedback for continuous improvement.

1.4 Non-Goals

The system must not:

- recommend candidates, parties, or ideologies
- persuade users whether to vote or abstain
- segment users politically
- generate campaign messaging
- give legal advice
- verify actual voter enrollment status unless linked to official status-checking resources
- collect or store personal voter details beyond what is minimally needed for session-level assistance
- replace official election authorities

1.5 Target Users / Personas

Persona	Description	Needs	Key UX Implications
First-Time Voter	18–22 years old, unfamiliar with process	Simple eligibility, registration, documents	Guided flows, plain language, onboarding
Rural Mobile User	Primarily uses low-end smartphone, may prefer voice	Regional language, voice support, step-by-step help	Lightweight UI, ASR/TTS, large buttons
Elderly Voter	May have accessibility or literacy limitations	Polling day guidance, assistance options	High contrast, large text, voice mode
Urban Professional	Wants quick answers	Fast FAQ, official links, concise summaries	Quick actions, source cards
Caregiver/Family Member	Helping another person vote	Document checklists, accessibility support	Structured flows, clear disclaimers
Journalist/Student	Wants process overview	Accurate explanations and references	Source-backed explainer content
Civic Volunteer	Needs neutral process education	Shareable, non-partisan procedural info	FAQ cards, official resource routing

1.6 Geographic Scope

In Scope

- India-wide election process education
- National-level election information concepts
- State-specific guidance where process or contacts differ
- Official resource routing to ECI and state CEO websites

Out of Scope

- Hyperlocal booth-level data unless sourced from official APIs or official links
- Predictive election analysis
- Campaign data or political strategy
- Candidate comparisons
- Opinion polling analysis

1.7 Compliance and Safety Boundaries

Mandatory Safety Principles

- Educational-only guidance
- No partisan persuasion
- No voter suppression or turnout manipulation
- No targeted political messaging
- No legal advice
- No fabricated deadlines or procedural certainty
- No collection of sensitive ID details unless operationally unavoidable and explicitly consented
- No retention of uploaded identity documents by default

Privacy Boundaries

- Do not ask for full EPIC number unless necessary for a linked official verification flow.
- Do not ask for Aadhaar number, full address, date of birth, or mobile number unless absolutely required by a user-initiated official form assistance scenario.
- If the user provides sensitive information voluntarily, redact it from logs where possible.
- Default analytics must be anonymized or pseudonymized.

1.8 Success Metrics and KPIs

Product KPIs

KPI	Definition	Target
Answer Helpfulness Rate	% of users rating response helpful	> 80%
Official Source CTR	% of sessions where users click official links when needed	> 35%
Low-Confidence Escalation Precision	% of low-confidence answers correctly escalated	> 90%
Language Task Completion	Completion rate of guided flows across all supported languages	> 75%
Voice Completion Rate	% of voice interactions completed successfully	> 65%
FAQ Deflection Rate	% of common queries handled without human support	> 70%
Safety Violation Rate	% of responses violating anti-partisanship/safety rules	< 0.1%
Hallucinated Fact Rate	% of sampled outputs with unsupported procedural claims	< 1%

Operational KPIs

KPI	Definition	Target
Median Response Time	Time from message submit to first answer	< 3.5 sec
P95 Response Time	95th percentile response time	< 8 sec
Translation Failure Rate	Failed translation calls / total	< 2%
ASR Accuracy Satisfaction	User-rated voice understanding accuracy	> 75%
Fallback Success Rate	Successful fallback when primary LLM/API fails	> 95%
Uptime	Service availability	> 99%

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2. Feature Specifications

2.1 Feature Summary Table

Feature	Priority	Description
Conversational Q&A	Must-have	Free-form questions about election process
Election Process Explainer Flows	Must-have	Guided educational pathways
Voter Registration Guidance	Must-have	Procedural registration help
Polling Day Guidance	Must-have	What to expect when voting
Eligibility Guidance	Must-have	Basic educational criteria
Document Checklist Helper	Must-have	Required document categories and reminders
State-Specific Information Routing	Must-have	Route users to relevant state guidance
FAQ Engine	Must-have	Fast responses for repeated questions
Misinformation Risk Handling	Must-have	Detect and safely respond to rumors
Escalation to Official Sources	Must-have	Link to official authorities when appropriate
Voice Input/Output	Should-have	Speech to text and text to speech
Session Memory Boundaries	Must-have	Minimal, privacy-respecting memory
Analytics/Admin Dashboard	Should-have	Product monitoring and content management

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2.2 Conversational Q&A

Purpose

Allow users to ask natural-language questions about election procedures in their preferred language.

Inputs

- user text
- optional voice-transcribed text
- selected language
- optional state selection
- session context
- device type metadata
- confidence/context signals

Outputs

- concise answer
- step-by-step instructions when applicable
- source cards
- disclaimers if relevant
- escalation links for official verification

Business Rules

- Answer only process-related election education queries.
- For personalized, legal, or uncertain scenarios, escalate to official sources.
- Use plain language first, then detail on demand.
- Always preserve neutrality.
- Cite sources when answering specific procedural claims.

Edge Cases

- User asks for political recommendation
- User asks if they should vote for a party
- User shares personal data
- User asks rumor-based question
- User asks unsupported state-specific detail

- User asks in mixed Hindi-English or transliterated form

Acceptance Criteria

- [ ] User can ask free-form questions in supported languages.
- [ ] Answers are non-partisan and procedural.
- [ ] Response includes official sources when details are date- or state-specific.
- [ ] Low-confidence cases trigger clarification or escalation.
- [ ] Sensitive user text is not echoed unnecessarily.

2.3 Election Process Explainer Flows

Purpose

Offer guided learning for users who do not know what to ask.

Inputs

- selected topic
- language
- state
- optional election type context

Outputs

- sequenced cards or chat steps
- definitions and process diagrams in text form
- next-step quick actions
- official source references

Recommended Explainer Topics

- How elections work in India
- Who can register as a voter
- How voter registration generally works
- What happens before polling day
- What to carry to the polling station
- How voting happens at the polling station
- How accessibility support works
- Common reasons for confusion or misinformation

Business Rules

- Use short modular steps.
- Offer “Tell me more” and “Open official source”.
- Avoid over-generalizing state-specific details.

Edge Cases

- User jumps mid-flow with a new question
- State-specific rules conflict with generic explainer
- User wants a summary instead of full explainer

Acceptance Criteria

- [ ] Each explainer can be completed in under 3 minutes on mobile.
- [ ] User can skip, repeat, or switch language.
- [ ] Relevant state-specific warnings appear where needed.

2.4 Voter Registration Guidance

Purpose

Explain general voter registration steps and route users to official registration portals or instructions.

Inputs

- age bracket
- citizenship confirmation prompt
- state/UT
- language
- whether user is first-time voter or updating details

Outputs

- procedural overview
- checklist of likely required information/documents
- official registration links/resources
- caution that final requirements should be confirmed on official portals

Business Rules

- Must not claim that a user is definitively eligible without caveats.
- Must distinguish between:
  - first-time registration

    - correction/update
    - address change
    - duplicate/transfer concerns
- Must instruct users to verify on official sources.

Edge Cases

- User is turning 18 soon
- User moved states recently
- User lacks standard proof documents
- User is overseas/NRI and asks about registration
- User asks for legal interpretation

Acceptance Criteria

- [ ] Provides clear step-by-step registration guidance.
- [ ] Identifies when official portal verification is needed.
- [ ] Avoids collecting unnecessary sensitive document numbers.

2.5 Polling Day Guidance

Purpose

Help users understand what happens on voting day.

Inputs

- state
- election type if available
- accessibility needs
- language

Outputs

- polling day checklist
- what to bring
- what to expect at polling station
- accessibility and assistance information
- link to official instructions

Business Rules

- Do not promise specific timings unless from official source.
- Clarify that exact instructions may vary by election and state.
- Separate “general process” from “confirm with official notice”.

Edge Cases

- User forgot voter ID
- User asks whether another ID can be used
- User needs disability assistance
- User asks about queue rules or companions
- User asks about voting if name missing from rolls

Acceptance Criteria

- [ ] Gives general polling workflow clearly.
- [ ] Offers official escalation when identity or roll inclusion must be confirmed.
- [ ] Includes accessibility support information when available.

2.6 Eligibility Guidance

Purpose

Provide educational explanation of basic voter eligibility concepts.

Inputs

- age
- citizenship self-declaration
- residence/state context
- language

Outputs

- simple explanation of basic eligibility principles
- what must be verified officially
- route to official resources

Business Rules

- Must not definitively adjudicate complex eligibility disputes.
- Use careful wording such as “generally,” “typically,” “please verify on official portal.”

Edge Cases

- User has recently moved
- User is NRI
- User is homeless or lacks common proof documents
- User asks about disqualification scenarios

Acceptance Criteria

- [ ] Explains basic eligibility accurately.
- [ ] Flags cases requiring official clarification.
- [ ] Never presents legal advice.

2.7 Document Checklist Helper

Purpose

Help users understand categories of documents they may need for registration or updates.

Inputs

- task type
- state
- language
- optional accessibility needs

Outputs

- categorized checklist
- plain-language explanation of why documents may be needed
- official source link
- reminders to confirm latest accepted document list

Business Rules

- Do not ask users to upload ID documents unless absolutely required and secure.
- Prefer showing categories rather than collecting document numbers.
- Clearly state that accepted documents may change.

Edge Cases

- User has no standard address proof
- User wants alternative documents
- State-specific accepted documents differ

Acceptance Criteria

- [ ] Checklist is task-specific and easy to read on mobile.
- [ ] Includes “confirm latest accepted documents” notice.
- [ ] Does not store document contents by default.

2.8 State-Specific Information Routing

Purpose

Route users to state/UT-specific instructions and official resources.

Inputs

- state/UT selector
- inferred state from query
- language

Outputs

- state-specific sources
- state CEO links
- customized cautions

Business Rules

- If state is unknown, ask a clarifying question.
- If no reliable state-specific content exists, provide general guidance plus official source escalation.
- Maintain a state metadata registry.

Edge Cases

- User refers to city only, not state
- User moved to another state
- State-specific source unavailable or down

Acceptance Criteria

- [ ] Agent asks for state when necessary.
- [ ] Correct official state resource displayed.
- [ ] Fallback to ECI/general guidance when state source unavailable.

2.9 FAQ Engine

Purpose

Instantly answer common questions with pre-reviewed content.

Inputs

- user query
- language
- query intent classification

Outputs

- FAQ answer
- source card
- suggestion chips for follow-up

Business Rules

- Use FAQ only when semantic similarity is above threshold.
- Pre-reviewed FAQ should take precedence over generative answers for common topics.
- Re-translate reviewed FAQ into target language where needed.

Edge Cases

- FAQ outdated
- Similar query but state-specific nuance required
- Mixed-language query

Acceptance Criteria

- [ ] Top FAQ topics answered within 1 sec where cached.
- [ ] FAQ entries are source-backed and versioned.
- [ ] User can request more detail from agent.

2.10 Misinformation Risk Handling

Purpose

Detect rumor-driven, manipulative, or potentially false election claims and respond safely.

Inputs

- user question
- confidence scores
- rumor keyword/semantic flags
- source availability

Outputs

- calm corrective response
- explanation of uncertainty
- official source links
- explicit note if claim cannot be verified

Business Rules

- Do not amplify rumors by restating them in sensational form.
- Use neutral wording.
- If claim cannot be confirmed, say so.
- Encourage verification using official sources.

Edge Cases

- Viral fake deadline
- False ID requirement claim
- Rumor about polling station procedures
- Manipulative prompts asking how to discourage voting

Acceptance Criteria

- [ ] Rumor-style queries trigger verification-first response.
- [ ] Manipulative or suppression-related requests are refused.
- [ ] Official source escalation included where relevant.

2.11 Escalation to Official Sources

Purpose

Send users to authoritative information when a claim is time-sensitive, state-specific, personalized, or high risk.

Inputs

- confidence score
- question type
- state
- language
- official resource registry

Outputs

- source cards
- deep links
- contact/helpdesk info where curated
- “what to ask/check” instructions

Business Rules

Escalation is mandatory when:

- answer depends on current deadlines
- answer depends on personalized registration status
- user seeks legal determination
- official state-level variation exists and cannot be confidently summarized
- the system detects uncertainty above threshold

Edge Cases

- Official website down
- User wants offline support
- User cannot read linked page language

Acceptance Criteria

- [ ] Relevant official source always available for escalated cases.
- [ ] System provides plain-language summary plus official next step.
- [ ] If source unavailable, fallback instructions are shown.

2.12 Voice Input/Output Support

Purpose

Support accessibility and low-literacy use cases.

Inputs

- microphone stream
- selected language
- text answer for TTS
- optional speaking speed preference

Outputs

- transcript
- agent response in text
- synthesized speech playback

Business Rules

- Align ASR/TTS language with UI language by default.
- Allow user correction of transcript before sending.
- Provide mute/stop/replay controls.
- Keep voice interactions optional.

Edge Cases

- background noise
- code-mixed speech
- unsupported dialect
- TTS unavailable in target language
- microphone permission denied

Acceptance Criteria

- [ ] User can speak question and review transcript.
- [ ] Agent can read answer aloud where TTS supported.
- [ ] TTS failure falls back to text-only gracefully.

2.13 Session Memory Boundaries and Privacy Behavior

Purpose

Maintain useful context without storing sensitive personal data.

Inputs

- prior messages
- selected language
- state

- flow progress

Outputs

- contextual continuity
- remembered preferences only
- privacy-safe session behavior

Business Rules

- Remember only:
  - selected language
  - selected state
  - recent topic context
  - accessibility preferences
- Do not persist:
  - identity document numbers
  - full addresses
  - Aadhaar numbers
  - date of birth
  - full legal names unless required and consented
- Time-box session memory.

Edge Cases

- User pastes a full ID number
- User requests deletion
- Shared device usage

Acceptance Criteria

- [ ] Session memory excludes sensitive identifiers by default.
- [ ] Clear session reset/delete option exists.
- [ ] Logs use redaction for obvious sensitive data patterns.

2.14 Analytics/Admin Dashboard Requirements

Purpose

Monitor quality, safety, usage, and content freshness.

Inputs

- anonymized analytics events
- user feedback
- moderation logs
- FAQ usage
- source click data

Outputs

- dashboard metrics
- content freshness alerts
- query cluster analysis
- safety incident reports

Business Rules

- Admin dashboard must not expose raw sensitive user data.
- PII redaction is required before analytics display.
- Provide state/language segmentation by aggregate only.

Edge Cases

- Surge in rumor queries
- Source links going stale
- Language-specific failure trends

Acceptance Criteria

- [ ] Dashboard shows top intents, satisfaction, source CTR, language breakdown.
- [ ] Admin can flag outdated FAQ/content.
- [ ] Safety and low-confidence trends are visible.

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3. Technical Architecture Using Free APIs

3.1 Architecture Principles

- Mobile-first, low-bandwidth aware
- API-abstraction layer to allow model/provider switching
- Official-source grounding first
- Privacy by design
- Language-aware orchestration
- Safe degradation when services fail
- Cache deterministic content aggressively

3.2 Core Service Roles

Service	Recommended Role
Antigravity Agent Platform	Agent orchestration, tool routing, prompt execution, workflow management
Gemini API	Primary reasoning, structured explanation, source-grounded synthesis
Groq API	Fast fallback generation, low-latency classification, summarization, intent detection
Bhashini	Translation, ASR, TTS, Indian language processing
Backend API	Session mgmt, state/resource lookup, moderation wrappers, persistence
Relational DB	Users, sessions, messages, resources, feedback, analytics references
Redis/Cache	FAQ caching, translation caching, session cache, rate-limited response cache
Object Storage	Optional audio files, logs export, content snapshots
Analytics Store	Event aggregation and dashboards

3.3 Recommended High-Level Architecture

```
Client Web App / Mobile Web
   |
   v
Backend API Gateway
   |
   +--> Auth / Session Service
   +--> Language Service
   +--> Resource Lookup Service
   +--> FAQ Service
   +--> Analytics Event Ingestion
   +--> Moderation / Safety Service
   |
   v
Antigravity Agent Orchestrator
   |
   +--> Tool: FAQ Search
   +--> Tool: Official Resource Resolver
   +--> Tool: State Metadata Lookup
   +--> Tool: Translation via Bhashini
   +--> Tool: ASR/TTS via Bhashini
   +--> Tool: LLM Primary via Gemini
   +--> Tool: LLM Fallback via Groq
   |
   v
Response Composer
   |
   v
Frontend UI
```

3.4 Proposed Service Usage Strategy

Gemini

Use for:

- primary answer generation
- reasoning over official content snippets
- multilingual explanation generation where supported
- structured content transformation
- uncertainty-aware response composition

Groq

Use for:

- low-latency fallback generation
- intent classification
- FAQ semantic routing
- response summarization
- moderation pre-check classification
- quick rephrasing in simple language

Bhashini

Use for:

- translation to/from Indian languages
- speech-to-text
- text-to-speech
- script-sensitive language handling
- voice alignment for regional language experience

3.5 Request Lifecycle

Text Query Lifecycle

1. User submits query.
2. Frontend sends:
    - message
    - selected language
    - state if selected
    - session ID
3. Backend performs:
    - input sanitation
    - rate limiting

    - sensitive-data pattern detection/redaction
    - language detection
4. FAQ classifier checks whether query matches reviewed FAQ.
5. If FAQ match is strong:
    - return FAQ answer + official source card.
6. Otherwise:
    - call moderation/risk classifier
    - enrich with state metadata
    - retrieve relevant official snippets/resources
    - invoke Antigravity agent with tools
7. Agent decides:
    - answer directly
    - ask clarification
    - escalate to official source
8. If primary LLM fails or times out:
    - retry once
    - fallback to Groq route
9. Backend stores sanitized transcript and events.
10. Frontend renders answer, source cards, feedback prompts.

Voice Query Lifecycle

1. User taps microphone.
2. Audio recorded locally.
3. Audio sent to backend or directly proxied to Bhashini ASR.
4. Transcript returned for confirmation.
5. User edits or accepts transcript.
6. Standard text lifecycle continues.
7. Final answer optionally sent to Bhashini TTS.
8. Audio URL returned to frontend for playback.

3.6 Fallback Strategy

Failure Condition	Fallback
Gemini unavailable	Use Groq for answer synthesis
Bhashini translation unavailable	Use Gemini/Groq translation fallback for text only
Bhashini ASR unavailable	Offer text input with friendly message
TTS unavailable	Text-only answer with replay disabled
State source unavailable	Use ECI/general guidance + state source unavailable message
FAQ DB unavailable	Route to live agent
Cache unavailable	Continue without cache
Analytics ingestion failure	Queue locally and retry async

3.7 Rate Limit and Resiliency Strategy

Controls

- IP + session rate limiting
- per-user request quotas
- burst control for voice/audio endpoints
- circuit breakers around third-party APIs
- async queue for analytics and audio processing
- retries with exponential backoff
- provider health checks and failover states

Suggested Policies

Layer	Policy
Chat endpoint	30 req / 5 min / session
Translate endpoint	60 req / 5 min / session
STT endpoint	10 uploads / 10 min / session
TTS endpoint	20 req / 10 min / session
Third-party retry	Max 2 retries with jitter
Timeout	8 sec primary LLM, 4 sec fallback classifier

3.8 Content Ingestion and Moderation Layer

Content Ingestion

- manually curated official sources list
- FAQ content seeded from reviewed procedural summaries
- periodic health checks on source URLs
- metadata tagging: national/state/topic/language/last-reviewed

Moderation Layer

- partisan intent detection
- suppression/manipulation detection
- legal advice detection
- misinformation/rumor detection
- self-check on confidence and source sufficiency

3.9 Recommended Stack

Layer	Recommendation
Frontend	Next.js / React / TypeScript
Backend	Node.js / Express or Fastify
Agent Orchestration	Antigravity Agent platform
Database	PostgreSQL
Cache	Redis
ORM	Prisma / Drizzle
Analytics	PostHog or self-hosted event pipeline
Hosting	Vercel/Netlify for frontend, Railway/Render/Fly.io for backend
Monitoring	Sentry + health checks

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4. UI/UX Design Specifications

4.1 Design Principles

- Black and white minimalist theme only
- High legibility
- Low cognitive load
- Mobile-first
- Fast interactions
- Accessibility-first
- Source transparency
- Trust through clarity, not decoration

4.2 Visual Design System

Color Palette

Token	Value	Usage
Background	#FFFFFF	Main background
Foreground	#000000	Primary text
Surface	#F7F7F7	Cards, panels
Border	#D9D9D9	Dividers, inputs
Muted Text	#5A5A5A	Secondary text
Inverse Background	#000000	Header/footer CTA blocks
Inverse Text	#FFFFFF	Text on black backgrounds
Error	#111111 with icon/text cue	Avoid red dependence; use text + icon
Focus Ring	#000000	Keyboard focus

Theme Rules

- No color-coded political associations
- No gradients
- No decorative illustrations required
- Emphasis through weight, spacing, borders, and typography

4.3 Typography

Element	Recommendation
Font Family	Inter, Noto Sans, and language-compatible Noto families
Body Size	16px minimum
Line Height	1.5–1.7
Headings	24/20/18 with strong weight
Button Text	14–16px semibold
Support for Scripts	Ensure compatible font fallback for Indic scripts

4.4 Spacing and Layout

- 8px spacing system
- Mobile padding: 16px
- Desktop container max-width: 960px
- Tap target minimum: 44x44px
- Chat bubbles full-width on mobile with generous padding
- Sticky bottom input bar for chat

4.5 Core Screens

1. Onboarding Screen

Elements:

- title
- trust statement
- educational-only disclaimer
- language selector
- optional state selector
- quick-start buttons

2. Main Chat Screen

Elements:

- top bar with language and state
- disclaimer pill/banner
- chat messages
- source cards
- quick action chips
- text box + mic button + send
- “Reset session” option

3. Guided Flow Screen / Chat-Integrated Cards

Elements:

- progress indicator
- step card
- short explanation
- next action buttons
- official source links

4. Feedback Modal

Elements:

- helpful / not helpful
- “report issue”
- issue category
- optional text

5. Admin Dashboard

Elements:

- top metrics
- language/state filters
- query clusters
- failed response logs
- source freshness alerts
- moderation incidents

4.6 UI Components Specification

Component	Behavior
Chat Bubble	User bubble white with black border; assistant bubble light-gray surface
Quick Action Chip	Rounded rectangle, black border, no color fill emphasis
Language Selector	Modal or dropdown with search and native script labels
Voice Controls	Mic, stop, replay, and transcript edit field
State Selector	Searchable dropdown with all states/UTs
Source Cards	Title, source label, last reviewed, open link CTA
Disclaimer Banner	Persistent but dismissible per session
Feedback Widget	Thumbs up/down + issue report
Empty State	Suggested prompts in selected language
Error State	Plain explanation + retry button + official source fallback

4.7 Accessibility Requirements

WCAG-Minded Requirements

- minimum contrast ratio compliant for text and controls
- visible keyboard focus
- semantic HTML
- ARIA labels on voice buttons and interactive controls
- screen-reader-friendly source cards
- support for text scaling up to 200%
- captions/transcripts for all voice output where feasible
- no information conveyed by color alone
- loading indicators announced to assistive tech

Accessibility-Specific Product Features

- “Read aloud” button
- “Simplify answer” button
- “Show step-by-step”
- “Large text mode”
- “High spacing mode”
- guidance for voters with disabilities routed to official resources

4.8 Onboarding UX Copy Requirements

Must clearly communicate:

- educational purpose
- non-partisan nature
- privacy-first behavior
- need to verify time-sensitive details from official sources

Example onboarding copy:

This assistant explains election processes in simple language. It does not support any party or candidate and may direct you to official election websites for confirmation.

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5. Multi-Language Support Requirements

5.1 Supported Languages

The platform should support the **22 scheduled languages of India**.

Language	ISO/Placeholder Code	Script	Notes
Assamese	as	Assamese/Bengali	Ensure script rendering
Bengali	bn	Bengali	Common in eastern states
Bodo	brx	Devanagari	Verify Bhashini coverage
Dogri	doi	Devanagari	Fallback may be needed
Gujarati	gu	Gujarati	Full support target
Hindi	hi	Devanagari	High-priority
Kannada	kn	Kannada	Full support target
Kashmiri	ks	Perso-Arabic / Devanagari variants	Clarify display support
Konkani	gom or kok	Devanagari primarily	Coverage validation needed
Maithili	mai	Devanagari	Coverage validation needed
Malayalam	ml	Malayalam	Full support target
Manipuri (Meitei)	mni	Meitei Mayek / Bengali	Define script support strategy
Marathi	mr	Devanagari	Full support target
Nepali	ne	Devanagari	Full support target
Odia	or	Odia	Full support target
Punjabi	pa	Gurmukhi	Full support target
Sanskrit	sa	Devanagari	Low-demand but included
Santali	sat	Ol Chiki	Coverage validation needed
Sindhi	sd	Arabic/Devanagari variants	Define fallback
Tamil	ta	Tamil	Full support target
Telugu	te	Telugu	Full support target
Urdu	ur	Perso-Arabic	RTL-like handling considerations for UI segments

5.2 Language Handling Rules

1. Detect input language automatically where possible.
2. Respect explicit user language selection over auto-detection.
3. If query is mixed-language, preserve user intent and answer in selected UI language unless user requests bilingual output.
4. If a language is not fully supported for generation or TTS:
    - provide text fallback
    - optionally provide Hindi or English secondary answer only if user consents
5. Keep source titles in original official language where needed, but provide translated summaries.

5.3 Translation Pipeline

Input-Side

- detect source language
- translate to orchestration language if needed
- preserve named entities
- preserve state names and official terms

Output-Side

- generate grounded answer
- translate to user language
- back-check critical terms
- attach source cards with translated summary and original source label

5.4 Script Display Considerations

- Use script-appropriate fonts
- Avoid letter spacing on Indic scripts
- Ensure line-height supports complex glyphs
- Test truncation carefully for longer scripts
- Do not romanize by default unless requested

5.5 Transliteration Considerations

The system should support:

- Hindi in Latin script (“mera naam voter list me kaise check karein?”)
- mixed Hindi-English
- state names written phonetically
- transliterated query normalization before intent classification

5.6 Mixed-Language Queries

Examples:

- “Voter ID ke liye documents kya chahiye?”

- “தமிழில் voter registration எப்படி?”
- “Name correction form ka process in Bengali”

Behavior:

- classify dominant language
- preserve intent
- answer in selected language or ask:
  - “Would you like the answer in Hindi / English / [selected language]?”

5.7 Voice-Language Alignment

- Default ASR/TTS language = current UI language
- If detected speech language differs from UI language, confirm switch
- TTS voice should match selected language when supported
- If TTS unavailable:
  - show transcript only
  - keep text visible for accessibility

5.8 Priority Support Tiers

Tier	Languages	Expectation
Tier 1	Hindi, English-adjacent support flows, Bengali, Tamil, Telugu, Marathi, Kannada, Malayalam, Gujarati, Punjabi, Odia, Urdu	Full text + preferred voice
Tier 2	Assamese, Nepali, Konkani, Maithili, Dogri, Sanskrit, Bodo	Strong text, voice where possible
Tier 3	Santali, Kashmiri, Sindhi, Manipuri	Fallback-aware support, validate coverage carefully

---

6. User Flows and Interaction Patterns

6.1 First-Time Onboarding Flow

Steps

1. User lands on homepage.
2. Sees trust statement and educational-only disclaimer.
3. Chooses language.
4. Optionally chooses state/UT.
5. Sees 4–6 quick-start actions:
   - Register to vote
   - Am I eligible?
   - What to carry on polling day?
   - Find official resources
   - Ask a question
6. Begins chat.

UX Notes

- Keep onboarding under 2 screens on mobile.
- No mandatory sign-in.
- State selection may be skipped and asked later.

6.2 Language Selection Flow

Steps

1. User taps language selector.
2. Searchable list opens with native script names.
3. User selects language.
4. Interface updates immediately.
5. Existing message history remains, but future answers switch language.
6. Offer “Translate previous answer” action.

UX Notes

- Remember language preference locally/session-level.
- Do not rely only on English labels.

6.3 Ask a Question Flow

Steps

1. User types or speaks a question.
2. System detects language and topic.
3. System checks FAQ and official-resource context.
4. Agent replies with:
   - short answer
   - step-by-step expansion
   - source card(s)
   - follow-up quick actions
5. User can ask follow-up questions.

UX Notes

- Show typing/loading state.
- Source cards should appear under relevant answer.
- “Was this helpful?” should appear after substantive answers.

6.4 Registration Help Flow

Steps

1. User selects “Register to vote.”
2. Agent asks:
    - Is this first-time registration or updating details?
    - Which state/UT?
    - Preferred language confirmed
3. Agent explains general registration process.
4. Agent provides document category checklist.
5. Agent shows official registration source links.
6. Agent asks if user wants:
    - simplified summary
    - checklist only
    - official link
    - how to correct details instead

UX Notes

- Avoid requesting exact personal details.
- Keep each step digestible.
- Use progress cues like “Step 1 of 4”.

6.5 Polling Day Help Flow

Steps

1. User selects “Polling day help.”
2. Agent asks for state if needed.
3. Agent provides:
    - what to carry
    - what happens at polling station
    - what if assistance is needed
    - what to do if there is confusion
4. Agent includes official source verification links.
5. User can request printable/shareable checklist view.

UX Notes

- Prioritize clarity and confidence.
- Offer “Read aloud” by default on this flow.

6.6 State-Specific Query Flow

Steps

1. User asks a state-dependent question.
2. Intent layer flags need for state.
3. Agent asks:
    - “Which state or union territory is this for?”
4. User selects from dropdown or types.
5. System fetches state metadata and official resources.
6. Agent answers with state-specific guidance or states limitations.

UX Notes

- Never guess the state without confirmation unless highly confident and easily reversible.
- If user says city only, infer possible state but confirm.

6.7 Voice Interaction Flow

Steps

1. User taps mic.
2. Permission prompt shown if first time.
3. User speaks.
4. Transcript appears in editable field.
5. User taps send or re-records.
6. Agent processes query.
7. Answer is shown in text and optionally read aloud.
8. User can pause, replay, or stop speech.

UX Notes

- Always show transcript for trust.
- Offer low-bandwidth mode with no TTS auto-play.

6.8 Official Source Escalation Flow

Steps

1. System detects high uncertainty or time-sensitive question.
2. Agent explains:
   - “This may vary by election or state.”
3. Agent provides:
   - brief general explanation
   - official source card
   - what to verify on the official page
4. User can open link or ask for a summary of source content.

UX Notes

- Never make escalation feel like a failure.
- Position it as a trust feature.

6.9 Low-Confidence Response Flow

Steps

1. Internal confidence score falls below threshold.
2. System avoids speculative answer.
3. Agent either:
   - asks a clarifying question
   - gives limited generic answer + official source
4. Logs low-confidence event.
5. Feedback prompt appears after answer.

UX Notes

- Prefer honest uncertainty over polished guesswork.
- Wording example:
   - “I may not have enough reliable detail to answer that accurately.”

6.10 Feedback / Report Issue Flow

Steps

1. User clicks “Not helpful” or “Report issue.”
2. Prompt asks issue category:
   - inaccurate information
   - unclear answer
   - language issue
   - source broken
   - safety concern
3. Optional free text entered.
4. Submission stored in feedback system.
5. User sees acknowledgment and optional retry suggestions.

UX Notes

- Keep under 3 taps.
- No forced sign-in to report issue.

---

7. System Prompts for the Agent

7.1 Production-Grade Master System Prompt

```
You are the Election Process Education Agent for India, deployed on Antigravity's Agent platform.

MISSION
Help users understand election-related civic processes in India in a clear, neutral, educational, and multilingual way. Your purpose

CORE BOUNDARIES
- You are strictly non-partisan.
- Do not persuade users to support or oppose any candidate, party, ideology, or political group.
- Do not help with campaign strategy, political targeting, turnout manipulation, voter suppression, or opinion shaping.
- Do not provide legal advice.
- Do not claim certainty when information may vary by state, election type, date, or official notification.
- Do not fabricate deadlines, accepted documents, eligibility conclusions, or procedural rules.
- Do not encourage users to bypass official procedures.
- Do not ask for or store sensitive voter data unless absolutely necessary for a user-requested official workflow, and even then mini

PRIVACY RULES
- Avoid requesting personal identifiers such as Aadhaar number, full EPIC number, full address, date of birth, or document numbers un
- Prefer giving category-level guidance rather than collecting personal details.
- If a user provides sensitive data, do not repeat it back unnecessarily. Summarize or redact when possible.

RESPONSE STYLE
- Use simple, clear, respectful language.
- Be concise first; provide more detail on request.
- Prefer step-by-step bullets for procedures.
- When relevant, separate:
  1) general guidance
  2) what to verify officially
- If the user seems confused, offer a simplified explanation.
```

```
- If the user requests voice-friendly content, use short sentences.

MULTILINGUAL BEHAVIOR
- Respond in the user's selected language when supported.
- If the query is mixed-language, infer intent and answer in the selected language unless the user requests bilingual output.
- Preserve official names, website names, forms, and state names carefully.
- If a language capability is limited, explain that and offer the closest supported alternative only with user permission.

SOURCE AND CITATION RULES
- Base specific procedural claims on curated official sources whenever available.
- Cite official sources in a user-friendly way using source cards or short references.
- For time-sensitive, personalized, or state-specific information, direct the user to official sources such as the Election Commissio
- If you lack sufficient reliable source support, say so clearly.

UNCERTAINTY HANDLING
- If confidence is low, do not guess.
- Ask a clarifying question if that could materially improve accuracy.
- Otherwise provide limited general information plus an official escalation path.
- Use phrases like:
  - "This may vary by state or election."
  - "Please verify this on the official election website."
  - "I do not have enough reliable information to confirm that."

MISINFORMATION HANDLING
- If a user asks about a rumor, respond calmly and neutrally.
- Do not amplify unverified claims.
- State whether the claim can be confirmed, contradicted, or not verified based on available official information.
- Encourage verification using official sources.
- Never generate sensational or speculative corrections.

SAFETY REFUSALS
Refuse requests that involve:
- supporting or opposing a candidate or party
- political persuasion
- voter suppression
- discouraging specific groups from voting
- targeted political messaging
- manipulation of election participation
When refusing, politely redirect to neutral civic-process information.

FORMATTING RULES
When answering procedural queries, use this structure where relevant:
1. Short answer
2. Steps
3. What to verify officially
4. Source(s)

For FAQ-style answers, use:
- Answer
- Important note
- Official source

For low-confidence cases, use:
- What I can say generally
- What you should verify
- Official source

ESCALATION RULES
Escalate to official sources when:
- the answer depends on current deadlines
- the answer is state-specific and not fully confirmed
- the user asks about personal voter registration status
- the user asks a legal/exception case
- official verification is necessary

ACCESSIBILITY RULES
- Mention accessibility options when polling-day or assistance questions arise.
- Use plain language for low-literacy users.
- Keep voice-friendly answers short and segmented.

OUTPUT GUARDRAILS
- Never present opinions as facts.
- Never say a user is definitely eligible or registered unless the user confirms this through an official source.
- Never invent accepted ID lists or deadlines.
- Never ask for document uploads by default.

If tools are available, prefer:
- official resource lookup
- FAQ lookup
- state metadata lookup
- translation or speech tools
before generating unsupported content.
```

7.2 Short Fallback Prompt

```
You are a neutral Election Process Education Agent for India. Provide only educational, non-partisan, procedural information. Do not
```

7.3 Example Tool/Routing Instructions for Antigravity

```
ROUTING POLICY

1. Run intent classification:
   - faq
   - registration
   - eligibility
   - polling_day
   - documents
   - state_specific
   - misinformation
   - prohibited_political
   - unclear

2. If prohibited_political:
```

```
   - refuse
   - redirect to neutral election process education

3. If faq confidence > threshold:
   - fetch FAQ entry
   - translate if required
   - respond with FAQ + source

4. If state-specific intent and no state in context:
   - ask user for state/UT
   - stop generation

5. If query needs official verification:
   - fetch official resources
   - provide concise general guidance + source cards

6. Use Gemini as primary generator for grounded synthesis.
7. If Gemini fails or exceeds timeout:
   - use Groq fallback for concise answer
   - maintain same safety and citation behavior

8. If input/output language differs from orchestration language:
   - use Bhashini translation
   - preserve names, forms, official titles

9. If voice input:
   - Bhashini ASR -> transcript confirmation -> normal flow

10. If voice output requested:
   - response text -> Bhashini TTS -> return audio URL

11. Log:
   - intent
   - language
   - state
   - confidence band
   - escalation flag
   - feedback event
Do not log unredacted sensitive identifiers.
```

---

8. Database Schema Recommendations

8.1 Design Principles

- relational core for consistency
- content versioning for official resources and FAQs
- PII minimization
- redact or hash sensitive patterns where possible
- separate analytics/event data from transactional data when volume grows

8.2 Core Tables

8.2.1 `users`

For optional anonymous or signed-in users.

Field	Type	Notes
id	UUID PK	
external_auth_id	varchar nullable	optional
is_anonymous	boolean	default true
created_at	timestamp	
updated_at	timestamp	
deleted_at	timestamp nullable	soft delete

8.2.2 `user_preferences`

Field	Type	Notes
id	UUID PK	
user_id	UUID FK users.id	
preferred_language	varchar	language code
preferred_state_code	varchar nullable	
accessibility_large_text	boolean	
accessibility_voice_default	boolean	
accessibility_high_spacing	boolean	
consent_analytics	boolean	
created_at	timestamp	
updated_at	timestamp	

8.2.3 `sessions`

Field	Type	Notes
id	UUID PK	
user_id	UUID FK users.id nullable	
session_token	varchar unique	
language_code	varchar	current session language
state_code	varchar nullable	

Field	Type	Notes
started_at	timestamp	
ended_at	timestamp nullable	
last_activity_at	timestamp	
channel	varchar	web/mobile-web
is_reset	boolean	
privacy_mode	boolean	optional enhanced privacy

8.2.4 `messages`

Field	Type	Notes
id	UUID PK	
session_id	UUID FK sessions.id	
role	varchar	user/assistant/system/tool
original_text	text	sanitized
translated_text	text nullable	
detected_language	varchar nullable	
response_confidence_band	varchar nullable	high/medium/low
escalation_flag	boolean	
faq_match_id	UUID nullable	
created_at	timestamp	

8.2.5 `message_redactions`

Field	Type	Notes
id	UUID PK	
message_id	UUID FK messages.id	
redaction_type	varchar	phone/id/address/etc
redacted_fragment_hash	varchar	avoid plain text storage
created_at	timestamp	

8.2.6 `languages`

Field	Type	Notes
code	varchar PK	
language_name_english	varchar	
language_name_native	varchar	
script	varchar	
translation_supported	boolean	
asr_supported	boolean	
tts_supported	boolean	
active	boolean	

8.2.7 `states`

Field	Type	Notes
code	varchar PK	state/UT code
name_english	varchar	
name_localized_json	jsonb	localized names
ceo_website_url	text	official source
helpdesk_contact	text nullable	curated
active	boolean	
updated_at	timestamp	

8.2.8 `official_resources`

Field	Type	Notes
id	UUID PK	
state_code	varchar FK states.code nullable	null for national
resource_type	varchar	portal/faq/contact/deadline/accessibility
title	varchar	
summary	text	reviewed summary
language_code	varchar nullable	source language
source_url	text	official URL
source_domain	varchar	validation
is_national	boolean	
last_reviewed_at	timestamp	
content_version	integer	
active	boolean	

8.2.9 `faq_entries`

Field	Type	Notes
id	UUID PK	
category	varchar	registration/polling/etc

Field	Type	Notes
canonical_question	text	
canonical_answer	text	reviewed
language_code	varchar	source language
translated_variants_json	jsonb	optional
source_resource_id	UUID FK official_resources.id	
embedding_ref	varchar nullable	if vector index external
active	boolean	
last_reviewed_at	timestamp	
version	integer	

8.2.10 `analytics_events`

Field	Type	Notes
id	UUID PK	
session_id	UUID FK sessions.id nullable	
user_id	UUID FK users.id nullable	
event_name	varchar	
event_properties	jsonb	
language_code	varchar nullable	
state_code	varchar nullable	
created_at	timestamp	

8.2.11 `feedback_reports`

Field	Type	Notes
id	UUID PK	
session_id	UUID FK sessions.id	
message_id	UUID FK messages.id nullable	
feedback_type	varchar	helpful/not_helpful/report
issue_category	varchar nullable	
comment	text nullable	sanitized
created_at	timestamp	
resolved_status	varchar	open/reviewed/resolved

8.2.12 `moderation_logs`

Field	Type	Notes
id	UUID PK	
session_id	UUID FK sessions.id	
message_id	UUID FK messages.id	
detected_label	varchar	rumor/political/legal/etc
severity	varchar	low/med/high
action_taken	varchar	answered/refused/escalated
model_used	varchar	
created_at	timestamp	

8.2.13 `provider_health_logs`

Field	Type	Notes
id	UUID PK	
provider_name	varchar	Gemini/Groq/Bhashini
endpoint_name	varchar	
status	varchar	ok/degraded/down
latency_ms	integer	
error_summary	text nullable	
created_at	timestamp	

8.3 Relationships

- `users` 1:N `sessions`
- `sessions` 1:N `messages`
- `messages` 1:N `message_redactions`
- `states` 1:N `official_resources`
- `official_resources` 1:N `faq_entries`
- `sessions` 1:N `analytics_events`
- `sessions` 1:N `feedback_reports`
- `messages` 1:N `moderation_logs`

8.4 Indexing Recommendations

Table	Index
sessions	`session_token`, `last_activity_at`
messages	`(session_id, created_at)`
messages	`response_confidence_band`
official_resources	`(state_code, resource_type, active)`

Table	Index
official_resources	`source_domain`
faq_entries	`(category, active)`
analytics_events	`(event_name, created_at)`
feedback_reports	`(resolved_status, created_at)`
moderation_logs	`(detected_label, severity, created_at)`

8.5 Retention Guidance

- Raw messages: 30–90 days depending on privacy policy
- Redaction hashes: 90 days max
- Analytics events: 12 months aggregated, raw shorter if needed
- Feedback reports: until resolved + audit retention
- Moderation logs: 6–12 months
- Audio files: delete rapidly, preferably within hours or after transcription/TTS delivery unless explicit user consent and operational need

---

9. API Integration Details

9.1 Backend API Principles

- All frontend calls go through backend proxy
- Do not expose third-party keys to client
- Return structured metadata for source cards and confidence flags
- Sanitize user input before persistence
- Version endpoints under `/api/v1`

9.2 Endpoint Summary

Endpoint	Method	Purpose
`/api/v1/chat`	POST	Main chat interaction
`/api/v1/translate`	POST	Text translation
`/api/v1/speech-to-text`	POST	Audio transcription
`/api/v1/text-to-speech`	POST	Audio synthesis
`/api/v1/resources`	GET	Official resources lookup
`/api/v1/faq/search`	GET/POST	FAQ retrieval
`/api/v1/feedback`	POST	User feedback
`/api/v1/analytics/events`	POST	Event ingestion
`/api/v1/session/reset`	POST	Reset session context
`/api/v1/health`	GET	Health check
`/api/v1/providers/health`	GET	Provider health summary

9.3 Chat Endpoint

Request

```
POST /api/v1/chat
{
  "sessionId": "8ec94f6a-b2b0-4c8b-a765-7c3d2c3ca777",
  "message": "What documents are usually needed for voter registration?",
  "language": "en",
  "stateCode": "KA",
  "inputMode": "text",
  "userPreferences": {
    "largeText": false,
    "voiceOutput": false
  }
}
```

Response

```
{
  "messageId": "b98d1db1-1edb-4b4c-b537-c9d7747c9911",
  "sessionId": "8ec94f6a-b2b0-4c8b-a765-7c3d2c3ca777",
  "answer": {
    "shortAnswer": "For voter registration, you may typically need identity and address-related supporting documents, depending on th
    "steps": [
      "Check whether you are registering for the first time or updating details.",
      "Keep basic identity and address-related documents ready, if required.",
      "Confirm the latest accepted documents on the official election website for your state or the national portal."
    ],
    "officialVerification": "Accepted document lists can change, so please verify the current requirements on the official election w
    "confidence": "medium",
    "escalationRequired": true
  },
  "sources": [
    {
      "title": "Election Commission of India",
      "type": "official",
      "url": "https://eci.gov.in",
      "label": "National official source"
    },
    {
      "title": "Chief Electoral Officer Karnataka",
      "type": "official",
      "url": "https://ceokarnataka.karnataka.gov.in",
      "label": "State official source"
```

```
    }
  ],
  "followUpSuggestions": [
    "Show a simple checklist",
    "How do I register for the first time?",
    "What if I changed my address?"
  ],
  "metadata": {
    "language": "en",
    "stateCode": "KA",
    "intent": "documents",
    "servedFromFaq": false
  }
}
```

9.4 Translate Endpoint

Request

```
POST /api/v1/translate
{
  "text": "What should I carry on polling day?",
  "sourceLanguage": "en",
  "targetLanguage": "hi"
}
```

Response

```
{
  "translatedText": "मतदान के दिन मुझे क्या साथ ले जाना चाहिए?",
  "provider": "bhashini",
  "sourceLanguage": "en",
  "targetLanguage": "hi"
}
```

9.5 Speech-to-Text Endpoint

Request

```
multipart/form-data
```

- `audioFile`
- `language`
- `sessionId`

Response

```
{
  "transcript": "मतदान के दिन क्या करना होता है",
  "language": "hi",
  "confidence": 0.88,
  "provider": "bhashini"
}
```

9.6 Text-to-Speech Endpoint

Request

```
POST /api/v1/text-to-speech
{
  "text": "मतदान केंद्र पर क्या करना है, मैं आपको चरणों में बताता हूँ।",
  "language": "hi",
  "voicePreference": "default_female",
  "speakingRate": 1.0
}
```

Response

```
{
  "audioUrl": "https://example-cdn.com/audio/tts/response-12345.mp3",
  "provider": "bhashini",
  "expiresInSeconds": 3600
}
```

9.7 Resources Lookup Endpoint

Request

```
GET /api/v1/resources?stateCode=MH&topic=registration&language=mr
```

Response

```
{
  "resources": [
    {
      "id": "res_001",
      "title": "Chief Electoral Officer Maharashtra",
      "summary": "Official state election information and voter services.",
      "url": "https://ceo.maharashtra.gov.in",
      "type": "state_official",
      "stateCode": "MH",
      "language": "mr",
      "lastReviewedAt": "2026-04-01T10:00:00Z"
    },
```

```
    {
      "id": "res_002",
      "title": "Election Commission of India",
      "summary": "National-level election information and official guidance.",
      "url": "https://eci.gov.in",
      "type": "national_official",
      "stateCode": null,
      "language": "en",
      "lastReviewedAt": "2026-04-01T10:00:00Z"
    }
  ]
}
```

9.8 Feedback Endpoint

Request

```
POST /api/v1/feedback
{
  "sessionId": "8ec94f6a-b2b0-4c8b-a765-7c3d2c3ca777",
  "messageId": "b98d1db1-1edb-4b4c-b537-c9d7747c9911",
  "feedbackType": "report",
  "issueCategory": "inaccurate_information",
  "comment": "The answer should mention that users must verify official document requirements."
}
```

Response

```
{
  "status": "accepted",
  "reportId": "fb_7890",
  "message": "Thank you. Your feedback has been recorded."
}
```

9.9 Analytics Event Endpoint

Request

```
POST /api/v1/analytics/events
{
  "sessionId": "8ec94f6a-b2b0-4c8b-a765-7c3d2c3ca777",
  "events": [
    {
      "eventName": "source_card_clicked",
      "properties": {
        "sourceType": "state_official",
        "language": "en",
        "stateCode": "KA"
      },
      "timestamp": "2026-04-21T10:30:00Z"
    }
  ]
}
```

Response

```
{
  "status": "ok",
  "ingested": 1
}
```

9.10 Health Check Endpoint

Response

```
{
  "status": "ok",
  "appVersion": "1.0.0",
  "timestamp": "2026-04-21T10:31:00Z",
  "dependencies": {
    "database": "ok",
    "redis": "ok",
    "gemini": "ok",
    "groq": "degraded",
    "bhashini": "ok"
  }
}
```

9.11 Authentication and Security Considerations

- Anonymous sessions allowed
- Optional admin authentication via OAuth/SAML/passwordless
- CSRF protection for browser sessions
- API keys stored server-side only
- Signed URLs for temporary audio delivery
- Input validation via schema library
- Rate limiting and abuse detection
- Audit logging for admin actions
- Encryption at rest for DB and object storage
- Strict CORS rules
- Content Security Policy enabled

---

10. Project Folder Structure

10.1 Recommended Monorepo Structure

```
election-process-education-agent/
├── apps/
│   ├── web/
│   │   ├── public/
│   │   ├── src/
│   │   │   ├── app/
│   │   │   ├── components/
│   │   │   ├── features/
│   │   │   ├── hooks/
│   │   │   ├── lib/
│   │   │   ├── styles/
│   │   │   ├── i18n/
│   │   │   └── types/
│   │   ├── tests/
│   │   └── package.json
│   └── admin/
│       ├── src/
│       ├── tests/
│       └── package.json
├── services/
│   ├── api/
│   │   ├── src/
│   │   │   ├── routes/
│   │   │   ├── controllers/
│   │   │   ├── services/
│   │   │   ├── providers/
│   │   │   ├── middleware/
│   │   │   ├── schemas/
│   │   │   ├── utils/
│   │   │   ├── prompts/
│   │   │   └── config/
│   │   ├── tests/
│   │   └── package.json
│   ├── worker/
│   │   ├── src/
│   │   └── package.json
│   └── content-ingestion/
│       ├── src/
│       └── package.json
├── packages/
│   ├── ui/
│   ├── types/
│   ├── config/
│   ├── localization/
│   └── analytics/
├── prompts/
│   ├── system/
│   ├── fallback/
│   ├── tools/
│   ├── evals/
│   └── guardrails/
├── data/
│   ├── seeds/
│   ├── faq/
│   ├── states/
│   └── official-resources/
├── db/
│   ├── migrations/
│   ├── schema/
│   ├── seed.ts
│   └── prisma/
├── tests/
│   ├── e2e/
│   ├── integration/
│   ├── performance/
│   ├── accessibility/
│   ├── multilingual/
│   └── safety/
├── docs/
│   ├── architecture/
│   ├── api/
│   ├── prompts/
│   ├── qa/
│   └── runbooks/
├── scripts/
│   ├── seed-data/
│   ├── health-checks/
│   └── content-review/
├── .github/
│   └── workflows/
├── package.json
├── turbo.json
├── pnpm-workspace.yaml
└── README.md
```

10.2 Folder Explanations

Folder	Purpose
`apps/web`	User-facing chat application
`apps/admin`	Internal analytics and content management dashboard
`services/api`	Main backend service
`services/worker`	Async jobs: analytics, audio cleanup, cache refresh
`services/content-ingestion`	Official resource sync, validation, review pipelines

Folder	Purpose
`packages/ui`	Shared black-and-white design system components
`packages/localization`	Language constants, labels, translation maps
`prompts`	System prompts, tool prompts, guardrails, eval prompts
`data/faq`	Reviewed FAQ content source files
`db`	Schema, migrations, seed scripts
`tests`	Cross-cutting test suites
`docs`	Architecture and operational documentation

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11. Official Sources and Data Strategy

11.1 Source-of-Truth Policy

The system must rely on **official election authority sources** as the preferred truth base.

Primary Source Categories

1. **Election Commission of India (ECI)**
2. **Official state Chief Electoral Officer (CEO) websites**
3. Official election notifications, manuals, FAQs, or voter service pages
4. Official portal pages for registration, status checks, and polling guidance

Example Source Domains

- https://eci.gov.in
- official state CEO websites such as:
  - https://ceodelhi.gov.in
  - https://ceokarnataka.karnataka.gov.in
  - other verified state domains as curated by operations team

11.2 Content Strategy

Curated Content Layers

- Layer 1: Manually reviewed canonical FAQs
- Layer 2: State metadata and official resource directory
- Layer 3: Grounded generation over verified source snippets
- Layer 4: Escalation links for personalized or current details

Content Freshness Rules

- Review high-impact topics monthly
- Review deadline-sensitive topics per election cycle
- Auto-flag stale resources if not reviewed in 30–60 days
- Run URL health checks daily

11.3 Source Validation Rules

- Only whitelist official domains
- Store source review date
- Track content owner/editor
- Flag resources changed since last review
- Avoid scraping dynamic or user-generated political content

11.4 Source Citation Behavior

For each specific answer:

- cite at least one official source when giving concrete procedural information
- if citing general concepts only, still offer official verification link
- show source label:
  - National Official Source
  - State Official Source
  - Reviewed FAQ based on Official Source

11.5 Data Minimization

Do not store:

- ID document images
- full personal addresses
- voter ID numbers unless absolutely necessary
- Aadhaar numbers
- exact DOB unless unavoidable and explicitly consented

If temporary handling is unavoidable:

- encrypt
- minimize retention
- delete immediately after task completion

---

12. Implementation Checklist

12.1 Phase 1: Discovery and Planning

Must-Have

- [ ] Confirm product scope and non-goals
- [ ] Define official source whitelist
- [ ] Identify top 100 election education intents
- [ ] Draft multilingual content and FAQ strategy
- [ ] Confirm legal/privacy review for educational-only positioning
- [ ] Define escalation policies and refusal policies

Nice-to-Have

- [ ] Conduct field interviews with first-time voters
- [ ] Validate voice UX with low-literacy users
- [ ] Create state-specific content prioritization roadmap

12.2 Phase 2: Project Setup

Must-Have

- [ ] Initialize monorepo
- [ ] Set up CI/CD
- [ ] Configure environment secret management
- [ ] Provision PostgreSQL and Redis
- [ ] Create schema migration pipeline
- [ ] Set up linting, formatting, and type checking

Nice-to-Have

- [ ] Storybook for UI components
- [ ] Preview environments per branch

12.3 Phase 3: Backend Development

Must-Have

- [ ] Build chat API
- [ ] Build resource lookup API
- [ ] Build feedback API
- [ ] Build analytics ingestion API
- [ ] Implement rate limiting
- [ ] Implement redaction middleware
- [ ] Add provider abstraction for Gemini/Groq/Bhashini

Nice-to-Have

- [ ] Async job queue
- [ ] provider performance comparison dashboards

12.4 Phase 4: Frontend Development

Must-Have

- [ ] Build onboarding
- [ ] Build chat UI
- [ ] Build language selector
- [ ] Build state selector
- [ ] Build source cards
- [ ] Build feedback widget
- [ ] Implement black-and-white theme
- [ ] Ensure mobile responsiveness

Nice-to-Have

- [ ] Progressive web app features
- [ ] Offline cached FAQ snippets

12.5 Phase 5: Agent Prompt Engineering

Must-Have

- [ ] Implement master system prompt
- [ ] Implement fallback prompt
- [ ] Implement routing logic
- [ ] Add anti-partisanship and misinformation guardrails
- [ ] Add low-confidence handling
- [ ] Add source citation formatting rules

Nice-to-Have

- [ ] Prompt A/B tests for answer clarity

- [ ] language-specific style tuning

12.6 Phase 6: Integrations

Must-Have

- [ ] Integrate Gemini
- [ ] Integrate Groq fallback
- [ ] Integrate Bhashini translation
- [ ] Integrate Bhashini ASR/TTS
- [ ] Implement official resource directory
- [ ] Seed FAQ database

Nice-to-Have

- [ ] Auto-refresh source metadata
- [ ] audio compression optimization

12.7 Phase 7: QA and Safety

Must-Have

- [ ] Functional testing
- [ ] Multilingual accuracy testing
- [ ] Safety refusal testing
- [ ] Accessibility testing
- [ ] API failure fallback testing
- [ ] Hallucination audits against official sources

Nice-to-Have

- [ ] Large-scale synthetic query evaluation
- [ ] crowd-based language review

12.8 Phase 8: Launch Preparation

Must-Have

- [ ] Production secrets configured
- [ ] Monitoring and alerts configured
- [ ] Admin dashboard ready
- [ ] Incident runbooks documented
- [ ] Feedback queue triage process defined

Nice-to-Have

- [ ] Public usage guide
- [ ] multilingual help center articles

12.9 Phase 9: Monitoring and Continuous Improvement

Must-Have

- [ ] Track top intents
- [ ] Track source CTR
- [ ] Review safety incidents weekly
- [ ] Refresh FAQs regularly
- [ ] Monitor provider uptime and failovers

Nice-to-Have

- [ ] intent clustering for new FAQ candidates
- [ ] user cohort analysis by language and state

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13. Testing Scenarios